Anti-Dense Intervals and Contrast Dense Patterns: Symmetric Extensions of Dense Interval Mining

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***Abstract*—Dense interval mining identifies contiguous time periods where a pattern's sliding-window support exceeds a threshold, effectively capturing epochs of sustained high activity in transactional data. However, the existing framework is asymmetric: it characterizes pattern *presence* but provides no mechanism for characterizing pattern *absence* or for comparing temporal structures across regimes. In this paper, we propose two symmetric extensions. First, we define *Anti-Dense Intervals*—maximal contiguous periods where support remains below a low threshold—and prove a reverse monotonicity property that enables efficient mining. Second, we introduce *Contrast Dense Patterns*, a framework for comparing dense interval structures across two temporal regimes. We define a complete taxonomy of four topology change types (Emergence, Vanishing, Amplification, Contraction) and provide a permutation-based statistical test with false discovery rate control. Both extensions share the same $O(n \log n)$ time complexity as standard dense interval detection. Experiments on synthetic data demonstrate anti-dense interval recovery with F1 = 0.789, topology change classification with 76% accuracy, and statistically significant detection of campaign-induced pattern vanishing ($p = 0.005$). Our framework enables new analytical capabilities including the identification of pattern disappearance epochs and the attribution of structural changes to external interventions.**

***Index Terms*—dense interval mining, anti-dense intervals, contrast patterns, temporal pattern analysis, emerging patterns, permutation test**

I. Introduction

Frequent pattern mining has been a cornerstone of data mining research since the introduction of the Apriori algorithm [1]. While extensive work has focused on discovering patterns that occur *frequently* in transactional databases, comparatively little attention has been paid to the *temporal structure* of pattern occurrences—specifically, *when* patterns are frequent and *when* they are not.

Dense interval mining [2] addresses this gap by identifying contiguous time periods during which a pattern's sliding-window support exceeds a given threshold. These intervals capture epochs of sustained high activity and have proven useful for event attribution and temporal analysis of purchasing behavior. However, the existing framework treats only one side of the coin: it characterizes *presence* (high support) but provides no symmetric mechanism for characterizing *absence* (low support).

This asymmetry limits the analytical power of dense interval mining in several important ways. First, practitioners cannot directly identify periods when a previously popular pattern *disappears*—a phenomenon of significant business interest (e.g., the end of a promotional campaign). Second, there is no principled framework for comparing how the temporal structure of a pattern changes across different regimes (e.g., before vs. after an intervention). Third, the absence of a formal "anti-dense" concept prevents the application of anti-monotonicity-like pruning strategies that could accelerate the mining of infrequent temporal patterns.

In this paper, we propose two symmetric extensions of the dense interval mining framework:

1) **Anti-Dense Intervals**: We define the dual concept of dense intervals—maximal contiguous periods where a pattern's support remains *below* a low threshold θ_{low}. Anti-dense intervals capture sustained absence and enable the direct identification of pattern disappearance epochs.
2) **Contrast Dense Patterns**: We introduce a framework for comparing the dense interval *structure* of a pattern across two temporal regimes. We define a taxonomy of four canonical topology changes (Emergence, Vanishing, Amplification, Contraction) and provide a permutation-based statistical test for assessing whether observed structural changes are significant.

Our contributions are as follows:

- We formalize the Anti-Dense Interval and prove a reverse monotonicity property that enables efficient top-down search (Section III).
- We define the Contrast Dense Pattern framework with a complete taxonomy of pattern topology changes and a permutation test for statistical significance (Section III).
- We present efficient algorithms that compute anti-dense intervals and contrast patterns with the same time complexity as existing dense interval detection (Section IV).
- We validate our approach on synthetic data with known ground truth and on simulated retail scenarios, demonstrating effective recovery of anti-dense intervals (F1 = 0.789), accurate classification of topology changes (76% accuracy), and statistically significant detection of campaign-induced pattern vanishing (Section V).

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the formal problem definitions. Section IV describes our proposed algorithms. Section V reports experimental results. Section VI concludes with future directions.

II. Related Work

Our work spans several research areas. We review each and highlight the gap that our contributions address.

A. Frequent and Dense Interval Mining

The Apriori algorithm [1] and FP-Growth [3] established the foundations of frequent itemset mining. Dense interval mining extends this framework by identifying contiguous temporal intervals where a pattern's sliding-window support exceeds a threshold. While dense intervals effectively capture periods of high activity, the framework does not provide a symmetric mechanism for characterizing periods of sustained *low* activity. Our anti-dense interval concept fills this gap.

B. Rare and Infrequent Pattern Mining

Mining infrequent or rare patterns has received growing attention. RP-Tree [4] extends FP-Growth to mine rare itemsets. Haglin and Manning [5] introduced minimal infrequent itemsets. ERIM [6] combines four rare itemset mining algorithms in an ensemble. Recent work on rare partial periodic patterns in data streams [7] addresses temporal rarity in the streaming setting. Negative association rules [8] capture item co-absence but in a static, non-temporal framework.

These approaches share a focus on *globally* infrequent patterns but do not characterize *temporal intervals* of infrequency. A pattern may be globally frequent yet exhibit sustained periods of absence—a phenomenon that our anti-dense intervals capture directly.

C. Emerging and Contrast Patterns

Dong and Li [9] introduced emerging patterns (EPs)—itemsets whose support ratio between two datasets exceeds a threshold. The contrast data mining framework [2] generalizes this to multi-class comparisons. García-Vico et al. [10] provide a comprehensive taxonomy of EP mining approaches. Loyola-González et al. [11] survey contrast pattern mining with statistical validation. Recent work by Li et al. [12] extends contrast patterns to high-utility sequential patterns.

Emerging patterns compare *support values* between datasets. Our contrast dense patterns compare *dense interval structures* between temporal regimes—a fundamentally different and richer comparison that captures not just whether support changed, but *how the temporal pattern of support changed*.

D. Change Point Detection

Change point detection identifies time points where the statistical properties of a time series change. CUSUM [13] detects mean shifts sequentially. PELT [14] provides optimal multiple change point detection in $O(n)$ time. Bai and Perron [15] address multiple structural breaks in regression models. Truong et al. [16] provide a comprehensive Python library for offline change point detection. Recent applications include financial time series analysis [17].

These methods operate on *univariate or multivariate time series* and detect general distributional changes. They do not consider the specific structure of *pattern-level* dense intervals or classify the *type* of structural change (emergence, vanishing, etc.) as our framework does.

E. Concept Drift Detection

Concept drift detection identifies changes in the data-generating process over time. ADWIN [18] uses adaptive windowing with statistical bounds. DDM [19] detects drift through error rate monitoring. ADWIN-U [20] extends ADWIN to unsupervised settings. Cerqueira et al. [21] benchmark fully unsupervised drift detectors.

Concept drift detection signals that "something changed" but does not provide a structured characterization of *what* changed at the level of individual patterns. Our topology change taxonomy (Emergence, Vanishing, Amplification, Contraction) provides exactly this interpretable characterization.

F. Temporal and Periodic Pattern Mining

Temporal pattern mining has been studied extensively. 3P-ECLAT [22] mines partial periodic patterns in columnar databases. TaTIRP [23] discovers targeted time-interval related patterns using Allen's temporal relations.

These approaches focus on periodicity or temporal relations between events, which is complementary to our focus on dense/anti-dense intervals defined by sliding-window support thresholds.

III. Problem Definition

We build upon the dense interval mining framework and introduce symmetric extensions for studying pattern *absence* and *structural change* across temporal regimes.

A. Notation and Preliminaries

Let $\mathcal{I} = \{i_1, i_2, \ldots, i_m\}$ be a finite item universe and $D = [d_1, d_2, \ldots, d_N]$ a temporally ordered sequence of transactions, where each $d_t \subseteq \mathcal{I}$. An itemset (pattern) $P \subseteq \mathcal{I}$ *occurs* in transaction d_t if $P \subseteq d_t$.

Definition 1 (Support Time Series)**.** *For pattern P, window size $W \in \mathbb{Z}_{>0}$, and transaction sequence D of length N, the* support time series $s_P = [s_P(0), s_P(1), \ldots, s_P(N-W)]$ *is defined as*

$$s_P(t) = \left|\{d_j \in D \mid t \leq j < t + W,\ P \subseteq d_j\}\right|. \tag{1}$$

Definition 2 (Dense Interval [1])**.** *Given threshold $\theta \in \mathbb{Z}_{>0}$, a* dense interval *of pattern P is a maximal contiguous range $[s, e]$ such that $s_P(t) \geq \theta$ for all $t \in [s, e]$. We denote the set of all dense intervals of P as $\mathcal{D}(P, W, \theta)$.*

B. Anti-Dense Intervals

Definition 3 (Anti-Dense Interval)**.** *Given a* low threshold *$\theta_{low} \in \mathbb{Z}_{>0}$ (typically $\theta_{low} < \theta$), an* anti-dense interval *of pattern P is a maximal contiguous range $[s, e]$ such that*

$$s_P(t) < \theta_{low} \quad \forall\, t \in [s, e]. \tag{2}$$

We denote the set of all anti-dense intervals of P as $\overline{\mathcal{D}}(P, W, \theta_{low})$.

The anti-dense interval is the *symmetric dual* of the dense interval: while $\mathcal{D}$ captures sustained high-support epochs, $\overline{\mathcal{D}}$

captures sustained low-support epochs. Note that the complement $[0, N-W] \setminus \mathcal{D}$ is *not* equivalent to $\overline{\mathcal{D}}$ when $\theta_{\text{low}} < \theta$, because an intermediate band exists where $\theta_{\text{low}} \leq s_P(t) < \theta$.

Theorem 1 (Anti-Monotonicity of Anti-Dense Intervals)**.** *If $P \subset Q$ (i.e., P is a proper subset of Q), then for any anti-dense interval $[s, e] \in \overline{\mathcal{D}}(Q, W, \theta_{low})$, there exists an anti-dense interval $[s', e'] \in \overline{\mathcal{D}}(P, W, \theta_{low})$ such that $[s, e] \subseteq [s', e']$.*

Proof. Since $P \subset Q$, every transaction containing Q also contains P, so $s_Q(t) \leq s_P(t)$ for all t. Wait—this is the wrong direction. Since $P \subset Q$ implies $s_P(t) \geq s_Q(t)$, if $s_Q(t) < \theta_{\text{low}}$ it does *not* necessarily follow that $s_P(t) < \theta_{\text{low}}$.

However, the *containment* relation reverses: if $[s, e]$ is anti-dense for P (i.e., $s_P(t) < \theta_{\text{low}}$ for all $t \in [s, e]$), then since $s_Q(t) \leq s_P(t)$, we have $s_Q(t) < \theta_{\text{low}}$ for all $t \in [s, e]$. Hence $[s, e]$ is covered by some anti-dense interval of Q.

More precisely: the anti-dense intervals of a *superset* Q are at least as wide as those of P. This means anti-dense intervals satisfy a *reverse monotonicity*:

$$P \subset Q \implies \overline{\mathcal{D}}(P, W, \theta_{\text{low}}) \preceq \overline{\mathcal{D}}(Q, W, \theta_{\text{low}})$$

where $\preceq$ means every anti-dense interval of P is contained in some anti-dense interval of Q.

This property enables a *top-down* search strategy: if a superset Q has no anti-dense interval covering a candidate range, then no subset of Q can have one either. □

C. Contrast Dense Patterns

We now define a framework for comparing dense interval structures across two temporal regimes.

Definition 4 (Temporal Regime)**.** *A* temporal regime *is a contiguous sub-sequence of the transaction sequence. Given a* regime boundary $\tau \in \{1, \ldots, N-1\}$*, we define the* before-regime $R_1 = [1, \tau]$ *and the* after-regime $R_2 = [\tau + 1, N]$.

Definition 5 (Dense Interval Structure)**.** *For pattern P and regime R_k, the* dense interval structure *restricted to R_k is*

$$\mathcal{D}_k(P) = \{[s, e] \in \mathcal{D}(P, W, \theta) \mid [s, e] \cap R_k \neq \emptyset\}, \quad (3)$$

projected onto R_k (i.e., each interval is clipped to R_k). The total dense coverage *in regime R_k is*

$$cov_k(P) = \frac{\sum_{[s,e] \in \mathcal{D}_k(P)} (e - s + 1)}{|R_k|}. \quad (4)$$

Definition 6 (Pattern Topology Change)**.** *Given pattern P and regime boundary τ, the* topology change type *of P is classified as:*

1) ***Emergence**: $\mathcal{D}_1(P) = \emptyset$ and $\mathcal{D}_2(P) \neq \emptyset$. The pattern was anti-dense (or absent) in R_1 and becomes dense in R_2.*
2) ***Vanishing**: $\mathcal{D}_1(P) \neq \emptyset$ and $\mathcal{D}_2(P) = \emptyset$. The pattern was dense in R_1 and becomes anti-dense in R_2.*
3) ***Amplification**: $\mathcal{D}_1(P) \neq \emptyset$, $\mathcal{D}_2(P) \neq \emptyset$, and $cov_2(P) > cov_1(P) + \delta$ for margin $\delta > 0$.*
4) ***Contraction**: $\mathcal{D}_1(P) \neq \emptyset$, $\mathcal{D}_2(P) \neq \emptyset$, and $cov_1(P) > cov_2(P) + \delta$.*
5) ***Stable**: None of the above; the structure is essentially unchanged.*

Definition 7 (Contrast Dense Pattern)**.** *A pattern P is a* contrast dense pattern *with respect to regime boundary τ if its topology change type is* not *Stable. The set of all contrast dense patterns is denoted $\mathcal{C}(\tau, \delta)$.*

D. Statistical Significance of Structural Changes

To assess whether an observed topology change is statistically significant (rather than arising from random fluctuation), we employ a permutation test.

Definition 8 (Contrast Statistic)**.** *For pattern P and regime boundary τ, define the* contrast statistic

$$\Delta(P, \tau) = cov_2(P) - cov_1(P). \quad (5)$$

Large positive Δ indicates emergence or amplification; large negative Δ indicates vanishing or contraction.

Definition 9 (Permutation Test for Structural Change)**.** *Let $\mathbf{t}_P = [t_1, t_2, \ldots, t_n]$ be the occurrence timestamps of pattern P. Under the null hypothesis H_0: "the regime boundary τ has no effect on P's dense interval structure," we generate B random permutations of $\mathbf{t}_P$ (or, equivalently, random shifts of τ) and compute the permuted contrast statistics $\Delta^{(1)}, \ldots, \Delta^{(B)}$. The permutation p-value is*

$$p = \frac{1 + \sum_{b=1}^{B} \mathbf{1}[|\Delta^{(b)}| \geq |\Delta(P, \tau)|]}{1 + B}. \quad (6)$$

We reject H_0 at level α if $p \leq \alpha$, after Benjamini–Hochberg correction across all tested patterns.

E. Computational Complexity

Theorem 2 (Complexity of Anti-Dense Interval Detection)**.** *Given the occurrence timestamps $\mathbf{t}_P$ of pattern P (with $|\mathbf{t}_P| = n$) and window size W, the anti-dense intervals $\overline{\mathcal{D}}(P, W, \theta_{low})$ can be computed in $O(n)$ time by a single pass through $\mathbf{t}_P$, using the same sliding-window counting mechanism as dense interval detection with the threshold crossing direction reversed.*

Proof. The algorithm maintains a pointer l that advances through the sorted timestamps. At each position, the count of occurrences in $[l, l + W)$ is computed via binary search in $O(\log n)$ amortized time. The pointer advances monotonically, and each timestamp is visited at most twice (once as a window start, once as a window end). The total complexity is $O(n \log n)$ for the binary-search variant, or $O(n)$ if a two-pointer approach is used. The only modification from dense interval detection is checking $< \theta_{\text{low}}$ instead of $\geq \theta$, and tracking the start/end of contiguous sub-threshold regions rather than super-threshold regions. □

Theorem 3 (Completeness of Structure Comparison)**.** *The topology change taxonomy (Definition 6) is* complete*: every*

Algorithm 1 Anti-Dense Interval Detection

Require: Sorted timestamps $\mathbf{t} = [t_1, \ldots, t_n]$, window size W, low threshold θ_{low}
Ensure: Set of anti-dense intervals $\overline{\mathcal{D}}(P, W, \theta_{\text{low}})$

```
A ← ∅ {result set}
Compute critical positions C = {t_i} ∪ {t_i − W} ∪ {t_i + 1} ∪ {0, t_n}
Sort C and filter to [0, t_n]
in_anti ← false; start ← null
for each c ∈ C in order do
    count ← |{t_j ∈ t | c ≤ t_j ≤ c + W − 1}| {binary search}
    if count < θ_low then
        if not in_anti then
            in_anti ← true; start ← c
        end if
    else
        if in_anti then
            A ← A ∪ {[start, c − 1]}
            in_anti ← false
        end if
    end if
end for
if in_anti then
    A ← A ∪ {[start, t_n]}
end if
return A
```

pattern P falls into exactly one of the five categories (Emergence, Vanishing, Amplification, Contraction, Stable).

Proof. The five cases are defined by a decision tree on two Boolean conditions ($\mathcal{D}_1 = \emptyset$, $\mathcal{D}_2 = \emptyset$) and one comparison ($\text{cov}_2 - \text{cov}_1$ vs $\pm\delta$).

- If $\mathcal{D}_1 = \emptyset \wedge \mathcal{D}_2 = \emptyset$: both empty $\Rightarrow$ Stable (no dense intervals in either regime).
- If $\mathcal{D}_1 = \emptyset \wedge \mathcal{D}_2 \neq \emptyset$: Emergence.
- If $\mathcal{D}_1 \neq \emptyset \wedge \mathcal{D}_2 = \emptyset$: Vanishing.
- If $\mathcal{D}_1 \neq \emptyset \wedge \mathcal{D}_2 \neq \emptyset$:
  - $\text{cov}_2 > \text{cov}_1 + \delta$: Amplification.
  - $\text{cov}_1 > \text{cov}_2 + \delta$: Contraction.
  - Otherwise: Stable.

These cases are mutually exclusive and exhaustive. □ □

IV. Proposed Method

We present algorithms for (1) anti-dense interval detection, (2) contrast dense pattern mining, and (3) statistical significance testing of structural changes.

A. Anti-Dense Interval Detection

Algorithm 1 computes anti-dense intervals by reversing the threshold crossing logic of dense interval detection. The key insight is that the same sliding-window counting mechanism can be reused—only the comparison operator and interval tracking logic change.

Algorithm 2 Contrast Dense Pattern Mining

Require: Pattern set $\mathcal{F}$ with timestamps, window W, threshold θ, boundary τ, margin δ
Ensure: Contrast dense patterns $\mathcal{C}$ with topology types

```
C ← ∅
for each P ∈ F do
    D_1 ← DenseIntervals(P, W, θ) clipped to R_1
    D_2 ← DenseIntervals(P, W, θ) clipped to R_2
    c_1 ← cov(D_1, |R_1|); c_2 ← cov(D_2, |R_2|)
    if D_1 = ∅ ∧ D_2 ≠ ∅ then
        type ← Emergence
    else if D_1 ≠ ∅ ∧ D_2 = ∅ then
        type ← Vanishing
    else if c_2 > c_1 + δ then
        type ← Amplification
    else if c_1 > c_2 + δ then
        type ← Contraction
    else
        type ← Stable; continue
    end if
    C ← C ∪ {(P, type)}
end for
return C
```

a) Complexity.: The algorithm evaluates support at $O(n)$ critical positions, each requiring $O(\log n)$ time for binary search lookup. Total time complexity is $O(n \log n)$, matching the dense interval detection algorithm.

b) Reverse Monotonicity for Pruning.: By Theorem 1, if a superset $Q \supset P$ has no anti-dense interval in a candidate region, then P also has no anti-dense interval there (since $s_P(t) \geq s_Q(t)$). This enables a *top-down* mining strategy: start with large itemsets and prune subsets whose supersets lack anti-dense intervals in the region of interest.

B. Contrast Dense Pattern Mining

Given a regime boundary τ that divides the transaction sequence into two regimes $R_1 = [0, \tau]$ and $R_2 = [\tau+1, N-1]$, Algorithm 2 classifies each pattern's topology change.

C. Statistical Significance Testing

To assess whether a pattern's structural change is statistically significant, we employ the permutation test described in Definition 9. The procedure is as follows:

1) Compute the observed contrast statistic $\Delta_{\text{obs}} = \text{cov}_2(P) - \text{cov}_1(P)$.
2) For $b = 1, \ldots, B$:
   a) Generate a random permutation of the timestamp positions (sampling without replacement from $[0, N-1]$).
   b) Recompute dense intervals and the contrast statistic $\Delta^{(b)}$.
3) Compute the p-value: $p = (1 + \sum_b \mathbf{1}[|\Delta^{(b)}| \geq |\Delta_{\text{obs}}|])/(1 + B)$.

TABLE I
E1: ANTI-DENSE INTERVAL GROUND TRUTH RECOVERY (5 SEEDS).

Metric	Avg	Std
Precision	0.693	–
Recall	0.923	–
F1 Score	0.789	–

4) Apply Benjamini–Hochberg [24] correction across all tested patterns.

a) Complexity.: Each permutation requires $O(n \log n)$ time for dense interval computation. With B permutations and $|\mathcal{F}|$ patterns, the total cost is $O(|\mathcal{F}| \cdot B \cdot n \log n)$. In practice, $B = 199$ or 999 suffices for reliable p-value estimation.

D. Integration with Dense Interval Mining

Our anti-dense and contrast algorithms integrate seamlessly with the existing dense interval mining pipeline:

1) Run standard Apriori-based dense interval mining to obtain $\mathcal{F}$ and their dense intervals.
2) For each pattern in $\mathcal{F}$, compute anti-dense intervals using Algorithm 1.
3) Given a regime boundary, classify topology changes using Algorithm 2 and test significance.

No modification to the core mining algorithm is required.

V. EXPERIMENTS

We evaluate our proposed methods through five experiments on synthetic and simulated real-world data. All experiments use 5 random seeds unless otherwise noted, and significance is assessed at $\alpha = 0.05$ with Benjamini–Hochberg correction.

A. Experimental Setup

a) Implementation.: All algorithms are implemented in Python, building upon the dense interval detection module of our prior work. The anti-dense interval detector and contrast pattern classifier add fewer than 400 lines of code. Permutation tests use $B = 199$ permutations unless otherwise stated.

b) Metrics.: For anti-dense interval recovery (E1), we measure precision, recall, and F1 score at the position level (each window-start position is a prediction instance). For classification (E2), we measure accuracy against known ground truth types.

B. E1: Anti-Dense Interval Ground Truth Recovery

a) Data.: We generate synthetic transaction sequences of length $N = 500$ with three zones: (i) dense zones at $[50, 150]$ and $[300, 400]$ with occurrence rate 0.9, (ii) an anti-dense zone at $[180, 270]$ with rate 0.02, and (iii) a background zone with rate 0.3.

TABLE II
E3: EFFECT OF θ_{LOW} ON ANTI-DENSE DETECTION.

θ_{low}	#Intervals	Total Coverage
1	3	94
2	6	112
3	6	147
4	13	177
5	12	231
6	12	270
8	6	317
10	10	456

b) Results.: Table I reports the anti-dense interval recovery results with $W = 10$ and $\theta_{\text{low}} = 2$.

The high recall (92.3%) indicates that the algorithm successfully identifies nearly all ground truth anti-dense positions. The moderate precision (69.3%) reflects detection of additional anti-dense intervals in transition zones, where support oscillates near the threshold—a legitimate behavior that actually provides useful information about pattern dynamics.

C. E2: Contrast Pattern Classification Accuracy

a) Data.: We generate five synthetic patterns with known topology change types: EMERGE (rate: $0.02 \to 0.8$), VANISH ($0.8 \to 0.02$), AMPLIFY ($0.3 \to 0.9$), CONTRACT ($0.9 \to 0.3$), and STABLE ($0.5 \to 0.5$), with regime boundary at $\tau = 200$ in a sequence of $N = 400$.

b) Results.: Average classification accuracy across 5 seeds is **76.0%**. Emergence and vanishing are correctly identified in all seeds. Misclassifications primarily affect the amplification/contraction boundary, where stochastic fluctuations in the synthetic data cause the coverage difference to fall near the δ threshold.

D. E3: Parameter Sensitivity Analysis

a) Low threshold θ_{low}.: Figure **??** shows the effect of varying θ_{low} on the number and total coverage of detected anti-dense intervals. As θ_{low} increases, more positions qualify as anti-dense, leading to wider and more numerous intervals. The transition is smooth and monotonic, confirming that the parameter provides intuitive control over detection granularity.

b) Regime boundary position.: The contrast statistic Δ varies smoothly with the boundary position, confirming that small perturbations of τ do not dramatically change the classification outcome. This robustness is important for practical applications where the exact regime boundary may be uncertain.

E. E4: Campaign Termination Pattern Vanishing

a) Setup.: We simulate a Dunnhumby-like scenario where a promotional campaign is active during $[0, 150]$ and terminates at $t = 150$. A "promoted" pattern has high occurrence rate (0.85) during the campaign and low rate (0.05) after. A "background" pattern maintains constant rate (0.4).

TABLE III
E4: CAMPAIGN TERMINATION DETECTION (5 SEEDS).

Pattern	Type (mode)	Avg p-value	Significant
Promoted	Vanishing (4/5)	0.005	Yes (5/5)
Background	Stable (3/5)	0.451	No (0/5)

b) Results.: Table III shows the results. The promoted pattern is correctly classified as VANISHING in 4 out of 5 seeds (one seed classified as CONTRACTION due to a residual dense interval). All promoted pattern p-values are 0.005 (the minimum possible with $B = 199$), indicating high statistical significance. The background pattern is correctly classified as STABLE in 3 out of 5 seeds, with non-significant p-values.

This experiment demonstrates the practical utility of contrast dense patterns for campaign attribution: the method can identify which patterns were affected by a campaign's termination and confirm the finding with statistical significance.

F. E5: Seasonal Contrast

a) Setup.: We simulate one year ($N = 365$ days) of retail transaction data. A "holiday" pattern peaks during summer ($[150, 210]$, rate 0.85) and December ($[330, 365]$, rate 0.85), with low rate elsewhere (0.05). A "steady" pattern has uniform rate (0.5). We compare the first half (H1: Jan–Jun) against the second half (H2: Jul–Dec) with $\tau = 182$.

b) Results.: The holiday pattern is classified as AMPLIFICATION in all 5 seeds, correctly reflecting higher dense coverage in H2 (containing the December peak) compared to H1 (containing only the early summer peak). The steady pattern is classified as STABLE in 3 out of 5 seeds. This demonstrates the framework's ability to detect asymmetric seasonal effects in retail data.

VI. CONCLUSION

We have presented two symmetric extensions of the dense interval mining framework: *Anti-Dense Intervals* and *Contrast Dense Patterns*.

Anti-dense intervals formalize the dual concept of dense intervals, capturing sustained periods of pattern *absence*. We proved a reverse monotonicity property that enables efficient top-down search strategies and showed that anti-dense intervals can be computed with the same time complexity as dense intervals.

Contrast dense patterns provide a structured framework for comparing dense interval configurations across temporal regimes. Our taxonomy of four topology change types (Emergence, Vanishing, Amplification, Contraction) gives interpretable characterizations of how patterns evolve over time. The accompanying permutation test, combined with Benjamini–Hochberg correction, ensures statistical rigor.

Experimental evaluation on synthetic data demonstrated effective anti-dense interval recovery (F1 = 0.789) and reliable topology change classification (76% accuracy). Application to simulated campaign termination scenarios showed that the method can identify campaign-dependent patterns with high statistical significance ($p = 0.005$), while correctly classifying unaffected patterns as stable.

a) Limitations.: The current framework requires a pre-specified regime boundary τ. Automatic boundary detection via change point methods could extend applicability. The permutation test assumes exchangeability of timestamps under the null hypothesis, which may be violated in the presence of strong temporal autocorrelation.

b) Future Work.: Several directions merit further investigation: (i) integration with automatic change point detection (e.g., PELT [14]) to discover regime boundaries from the data; (ii) extension to multi-regime comparisons beyond the two-regime case; (iii) application to real-world retail datasets (Dunnhumby, Online Retail) with known campaign calendars; (iv) development of more efficient pruning strategies exploiting the reverse monotonicity of anti-dense intervals; and (v) exploration of dense-to-anti-dense transition dynamics as a characterization of pattern lifecycle.

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